AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2017

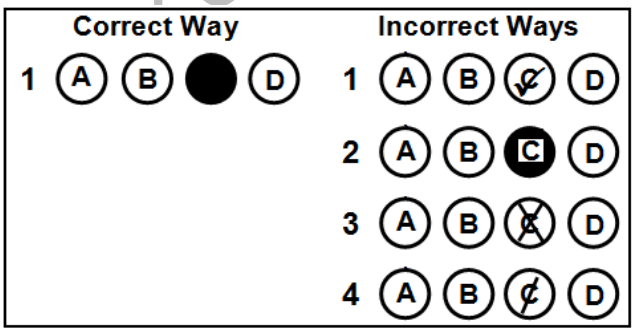
Chemistry Paper I

Time: 40 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. The total number of atoms of hydrogen and oxygen in 6.02×10^{23} molecules of water is

   A. 6.02×10^{23}
   B. 12.04×10^{23}
   C. 18.06×10^{23}
   D. 24.02×10^{23}

2. Which of the following has maximum number of moles?

   (**Note**: Atomic mass of C = 12 amu, H = 1 amu, O = 16 amu, Cl = 35.5 amu and Na = 23 amu)

   A. 20 g of HCl
   B. 20 g of H_2O
   C. 20 g of C_6H_6
   D. 20 g of NaCl

3. The amount of products obtained in a chemical reaction is called the

   A. total yield.
   B. actual yield.
   C. theoretical yield.
   D. percentage yield.

4. Which of the following metals has seven electrons in its 3d sub shell in the +2 oxidation state?

   A. $_{24}Cr$
   B. $_{26}Fe$
   C. $_{27}Co$
   D. $_{29}Cu$

5. In a discharge tube experiment, the rays which pass through the perforated cathode and produce a glow on the wall opposite to the anode are

   A. X-rays.
   B. gamma rays.
   C. cathode rays.
   D. positive rays.

6. According to Bohr's atomic model, when an electron jumps from a lower to higher orbit, it

   A. absorbs energy.
   B. releases energy.
   C. drops into the nucleus.
   D. changes into a neutron.

7. Which of the following pairs will NOT form a buffer solution?

   A. NH_4OH and NH_4Cl
   B. HCOOH and HCOONa
   C. NH_4OH and CH_3COONH_4
   D. CH_3COOH and CH_3COONa

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8. The molarity of 0.25 dm^3 of a solution containing 10.35 g of K_2CO_3 is

(**Note**: Atomic mass of C = 12 amu, O = 16 amu, K = 39 amu)

A. 0.075 M
B. 0.30 M
C. 2.58 M
D. 41.40 M

9. An acidic buffer is prepared by mixing a weak acid and a salt of

A. weak acid with a strong acid.
B. weak base with a strong acid.
C. weak acid with a strong base.
D. weak base with a strong base.

10. Which of the following molecules has a zero dipole moment?

A. HF
B. H_2O
C. CH_4
D. NH_3

11. The correct order for the decrease in repulsion between electron pairs is

A. lone pair-lone pair > lone pair-bond pair > bond pair-bond pair
B. lone pair-lone pair > bond pair-bond pair > lone pair-bond pair
C. bond pair-bond pair > lone pair-bond pair > lone pair-lone pair
D. lone pair-bond pair > lone pair-lone pair > bond pair-bond pair

12. If the electronegativity difference increases between hydrogen and halogen in a hydrogen halide molecule, then what will be the effect on its ionic character and bond strength?

	Ionic Character	**Bond Strength**
A	Increases	Increases
B	Increases	Decreases
C	Decreases	Increases
D	Decreases	Decreases

13. At which of the following conditions, does a real gas deviate from an ideal behaviour?

	Temperature	**Pressure**
A	Low	High
B	High	Low
C	High	High
D	Low	Low

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14. In a closed system, at a constant temperature, an increase in volume leads to a decrease in the pressure of the gas, because the

    A. space between the molecules decreases.
    B. average kinetic energy of the molecules increases.
    C. force of attraction between the molecules increases.
    D. collision of molecules with the walls of the container decreases.

15. Heat of vapourisation of a liquid is much greater than the heat of fusion of solid. Which of the following changes occurs during heat of vapourisation of liquid?

	Intermolecular Distance	**Kinetic Energy**
A	Decreases	Decreases
B	Decreases	Increases
C	Increases	Decreases
D	Increases	Increases

16. When water is poured into a glass test tube, the surface of the water is concaved. This meniscus forms because of

    A. weak adhesive forces.
    B. strong adhesive forces.
    C. strong cohesive forces.
    D. equal adhesive and cohesive forces.

17. In solid state, the molecules of iodine align in the form of layer lattice. This is an example of a/ an

    A. ionic solid.
    B. metallic solid.
    C. covalent solid.
    D. molecular solid.

18. The crystalline substance which shows variation in physical properties in different directions is known as

    A. symmetric.
    B. anisotropic.
    C. isomorphic.
    D. polymorphic.

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19. The ionisation of $KClO_3$ in the given reaction can be suppressed by adding

$$KClO_{3(s)} \rightleftharpoons K^{+}_{(aq)} + ClO^{-}_{3(aq)}$$

A. KCl
B. HCl
C. NaCl
D. NH_4Cl

20. What will be the effect of the increase in pressure on the given reaction at equilibrium?

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$$

A. It will remain constant.
B. It will stop spontaneously.
C. It will move in the forward direction.
D. It will move in the backward direction.

21. Which of the following statements is FALSE about exothermic reaction?

A. Temperature of the system increases with the release of energy.
B. Products obtained are at a lower energy level than the reactants.
C. A continuous source of energy is required to complete the reaction.
D. Energy of activation of forward reaction is less than that of backward reaction.

22. The rate of a chemical reaction can be retarded by adding a/ an

A. indicator.
B. activator.
C. inhibitor.
D. co-factor.

23. What is the molality of a solution prepared by dissolving 148 g of butanol ($C_4H_{10}O$) in 0.1 kg of benzene?

(**Note**: Atomic mass of C = 12 amu, O = 16 amu, H = 1 amu)

A. 20 mol/kg
B. 2 mol/kg
C. 0.2 mol/kg
D. 0.05 mol/kg

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24. Toothpaste is an example of a colloidal solution because its particles

A. can scatter a beam of light.
B. can be seen with the naked eye.
C. can form a homogenous mixture.
D. cannot pass through a filter paper.

25. All of the following statements define Raoult's law EXCEPT

A. relative lowering of vapour pressure depends on temperature.
B. relative lowering of vapour pressure is equal to the mole fraction of solute.
C. lowering of vapour pressure is directly proportional to the mole fraction of solute.
D. the vapour pressure of solution containing non-volatile solute is directly proportional to the mole fraction of solvent.

26. Below are the thermochemical reactions for the oxidation of carbon and carbon monoxide with the associated enthalpy changes.

$$C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)} \qquad \Delta H = ?$$

$$C_{(s)} + \tfrac{1}{2}O_{2(g)} \longrightarrow CO_{(g)} \qquad \Delta H_1 = -109\ \text{kJmol}^{-1}$$

$$CO_{(g)} + \tfrac{1}{2}O_{2(g)} \longrightarrow CO_{2(g)} \qquad \Delta H_2 = -284.5\ \text{kJmol}^{-1}$$

Based on the given information, the enthalpy of combustion for carbon forming carbon dioxide will be

A. $+393.5\ \text{kJmol}^{-1}$
B. $-393.5\ \text{kJmol}^{-1}$
C. $+175.5\ \text{kJmol}^{-1}$
D. $-175.5\ \text{kJmol}^{-1}$

27. A non-spontaneous reaction occurs in

A. dissolution of ionic salt in a polar solvent.
B. formation of nitric oxide in air by lightning.
C. burning of methane gas in the presence of air.
D. neutralisation of a strong acid by a strong base.

28. The expression used for molar heat capacity is

A. Heat / ΔT
B. Heat/ gram $\times \Delta T$
C. Heat / moles $\times \Delta T$
D. Heat $\times \Delta T$ / moles

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29. The oxidation number of sulphur in SO_3^{2-} is

A. + 2
B. + 3
C. + 4
D. + 6

30. During corrosion, water layer present on the surface of iron can dissolve

I. O_2
II. CO_2
III. N_2

A. I only
B. III only
C. I and II
D. II and III

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END OF PAPER

Please use this page for rough work

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